

G1 Campaign 2 — Deployment & Test Protocol (v2)

Adrián Lendínez Ibáñez — University of Bedfordshire, 22 July 2026. v2 after the pre-session audit (30 findings, 22 adversarially verified): adds the door-zone deterministic floor, spill-marker v2 with heartbeat/weights/invalid-run commands, machine-readable fill and per-run config snapshot, furniture census safeguards, vision watchdog, and the corrected statistics plan. Supersedes v1.

0. One-time deployment (GPUEDGE)

```
cd ~/Documents/G1_UNITREE_ROBOT_META_REASONING && git pull --rebase origin main
# GPU vision verified 21-Jul (72/72 bins, metric ranges). Launch (note cy=90 and --debug):
G1_FLOORCOLOR=1 python perception_server.py --host 0.0.0.0 --port 8008 \
--fx 300 --fy 300 --cx 160 --cy 90 --cam-h 1.10 --cam-pitch -10 --debug
# live viewer: http://192.168.51.80:8008/view · health: curl -s localhost:8008/health
```

1. Session start ritual (EVERY robot boot)

- Robot boots → iPhone Bluetooth ON → connect FROM the app (rotating Wi-Fi key) → teleop works in-app.
- USB → ios_webkit_debug_proxy → optional injection check: python3 g1_inject_teleop.py sniff then drive (W/S/A/D, SPACE stop, X exit).
- **Spill marker v2** (leave running ALL session): python spill_mark.py — ENTER = one spill (a slosh burst = ONE mark); w 200 = record cup weight; i reason = mark current run INVALID. It heartbeats every 2 s: the dataset now records gt_hb_seen/gt_alive_at_end/gt_dropouts, and the run console alarms if the marker dies mid-run.
- python3 g1_goto.py noisecheck (20 s standing) → walk near a waypoint → python3 g1_goto.py reloccheck.

2. Furniture census (REQUIRED once, before water)

```
# robot standing 1-1.5 m facing the sofa arm (NO people in the camera view):
G1_PERC=127.0.0.1:8008 python3 g1_goto.py furniture 15
# it lists candidate cells and asks for CONFIRMATION before writing (s/N).
# Check the printed positions fall ON the item. Repeat from a 2nd angle; then the low cabinet.
# Mistake? -> python3 g1_goto.py furniture undo · (restores the pre-census backup)
git add nav_map.json && git commit -m "furniture census" && git push
```

3. Dry shakedown (2 runs, no water) — already PASSED 21-Jul; repeat only after big changes

Checklist: reaches both ways · door crossed clean (door_center event with gap≈0.85) · perc_n in the dozens · no !C stops in open space · meta2_pf/meta2_unc/sent in samples. NEW since v1: door_guard events near the vano are the deterministic floor engaging — expected, not a fault.

4. Data runs

Water protocol: weigh BEFORE and AFTER every leg and record it in the marker (w 200 / w 153 — it lands in the dataset as weigh events); never refill between legs of a pair; a burst = ONE mark; governed HELP stop = valid result; marker dies mid-run → run INVALID (i marker died), redo. Every run command now carries G1_ARM and G1_FILL_G — they go into the dataset (arm, payload, full env_g1 + git SHA snapshot).

C2-GOV	6 (3 pairs B/A)	200 g	headline governed arm; keep state file across runs
C2-BASE	6 (3 pairs) **interleaved ABBA with C2-GOV**	200 g	n=4 made round-trip p<0.05 unreachable; 6 + ABBA kills drift confounds
C2-FULLCUP	2 gov + 2 base	247 g	the headline: campaign-1 governed aborted 4/4 here
C2-WRONG	6	200 g	wrong-prior recovery; fresh meta2_state_c2_wrong.json; write PI after each run
C2-LID	4 (2+2)	200 g	sealed lid; also PF validation
C2-ABL	2 (optional)	200 g	G1_COLORBRAKE=0 G1_DOORSTICKY=0 G1_DOORGUARD=0 ablation

Governed command (C2-GOV): — note the fresh state name c2b (the old c2_m2 carried an invalid-run door fail and was retired):

```
G1_ARM=C2-GOV G1_FILL_G=200 G1_CUP_G=246 \
G1_META2=2 G1_M2_CFG=config_meta2_g1payload.json G1_M2_REALCAL=1 \
G1_M2_STATE=meta2_state_c2b_m2.json G1_M2_STATE_INIT=meta2_state_twin_explored.json \
G1_M2_L3=1 G1_M2_L4=1 G1_M2_PROFILES=1 G1_M2_DOORLIB=1 G1_M2_FRAGSPEED=1 \
G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B # alternate: gotoviz A
```

Baseline command (C2-BASE):

```
G1_ARM=C2-BASE G1_FILL_G=200 G1_CUP_G=246 \
G1_META2=0 G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B
```

Optional slew pairs (Ext E, after the base pairs): add G1_SLEW=1 to the governed command for 2 pairs and compare marks/g per leg.

5. After every block

```
git pull --rebase origin main
git add dataset/ runs_summary.csv meta2_state_c2b_*.json goto.log nav_map.json
git commit -m "C2-GOV pair 2: 200g, B+A, weights 200->161->148"
git push
```

6. What changed under the hood (v2, all reversible)

DOOR-GUARD: fwd ≤0.28, turn ≤0.38, body ≥0.28 (non-aggressive) within 1.2 m of the vano, outside negotiation phases	21-Jul: marker died → fragility asleep → Efficient hit the frame; deterministic floor below the reasoner	G1_DOORGUARD=0
DOOR-CENTER v2: static-evidence cells only + median-of-5 + single-jamb warning	audit: one spurious live cell could shift the measured centre ~12 cm and pass the sanity check	G1_DOOR_CENTER=0
L2 veto monotonicity: vetoing the conservative can never deploy a riskier analogy (retains + clamps cap 0.24)	the 14-Jul incident, now impossible by construction	(bridge, always on)
Marker v2 heartbeat + weigh/invalid commands	"zero spills" vs "marker dead" now distinguishable; grams machine-readable	—
Vision hot watchdog (alarm + events if the channel dies >15 s mid-run)	21-Jul: channel died 62 s mid-run silently	—
Turn-cap only in cruise phases; SLEW exempts unstick; FRAGSPEED REALCAL anchor (1 spill now brakes ~3%, 2 ~12%)	audit fixes	—
Per-run config snapshot (env G1_*, git SHA, arm) + payload grams in the dataset	paper table must not depend on human memory	—